1. Monte Carlo → offline

E2 → A $\xrightarrow{a}$ B $Q(A,a) = 10$

[4,10,20,...] ↓ average diff episode

Temporal Diff. TD(0)

$V(S_t) \leftarrow V(S_t) + \alpha((G_t) - V(S_t))$ S_t = non terminal state

→ Return → After episode was completed.

$Q(A,a)+4$

E1 → A $\xrightarrow[a_{t+1}]{R_{t+1}}$ (B) $\xrightarrow[a_{t2}]{R_{t+2}}$ C $\xrightarrow[a_{t3}]{R_{t+3}}$ D $\xrightarrow[a_{t+4}]{R_{t+4}}$ Terminal $R_{t+4} = 10$

$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$

$= R_{t+1} + \gamma(R_{t+2} + \gamma R_{t+3} + \dots)$ $G_t = R_{t+1} + \gamma(\ )$

$\boxed{G_t = R_{t+1} + \gamma G_{t+1}}$ → Recursion ⇒ Start from last

G_T ← Terminal state return → Back Track

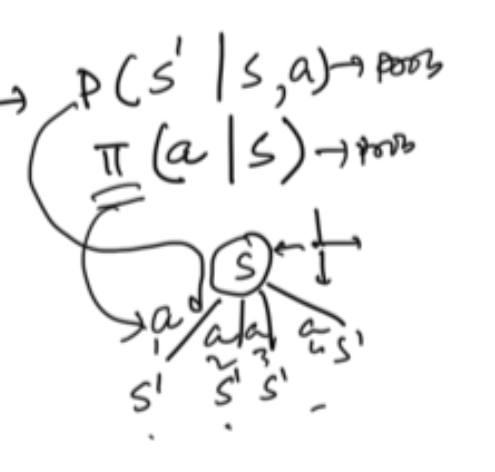
$\gamma G_{T-1}, G_{T-2}, G_{T-3}, \dots$

2. On policy MC.

Initialization
1. Initialize: π ← an arbitrary policy
2. $Q(S,A) \in \mathbb{R}$ → arbitrary Initialization.
3. Return (S,a) ← Empty set

GPI → Generic Policy Iteration
1. Policy evaluation
2. Policy Improvement

DP → Planning Based on Probabilities → Stochastic → $P(s'|s,a)$ → prob, $\pi(a|s)$ → prob



Sampling

Repeat forever

1. Generate an episode π
   $S_0, A_0, R_1 \rightarrow S_1, A_1, R_2 \rightarrow S_2, A_2, R_3 \rightarrow \dots$
2. $G \leftarrow 0$
3. Loop for each step of the episode, from the last non-Terminal state (T) → Offline.
   $G \leftarrow R_{t+1} + \gamma G$ → (Actual Returns from the past)
   unless pair (S_t, A_t) found
   1. append $G_t(S,a)$ to Return (S_t, A_t) [ ]
   2. $\boxed{Q(S_t, A_t) \leftarrow \text{average}(\text{Return}(S_t, A_t))}$

   Policy is determined indirectly from Q.
   3. $A^* \leftarrow \arg\max_a Q(S_t, a)$ → greed action

Retrospection

E1 = A→B→C→A→B→(F→T)

E2 = A→B→C→(F→T)

S	a1	a2	a3	a4
S0	r1 r2	r2, r3	r3, r4	

A--B--C

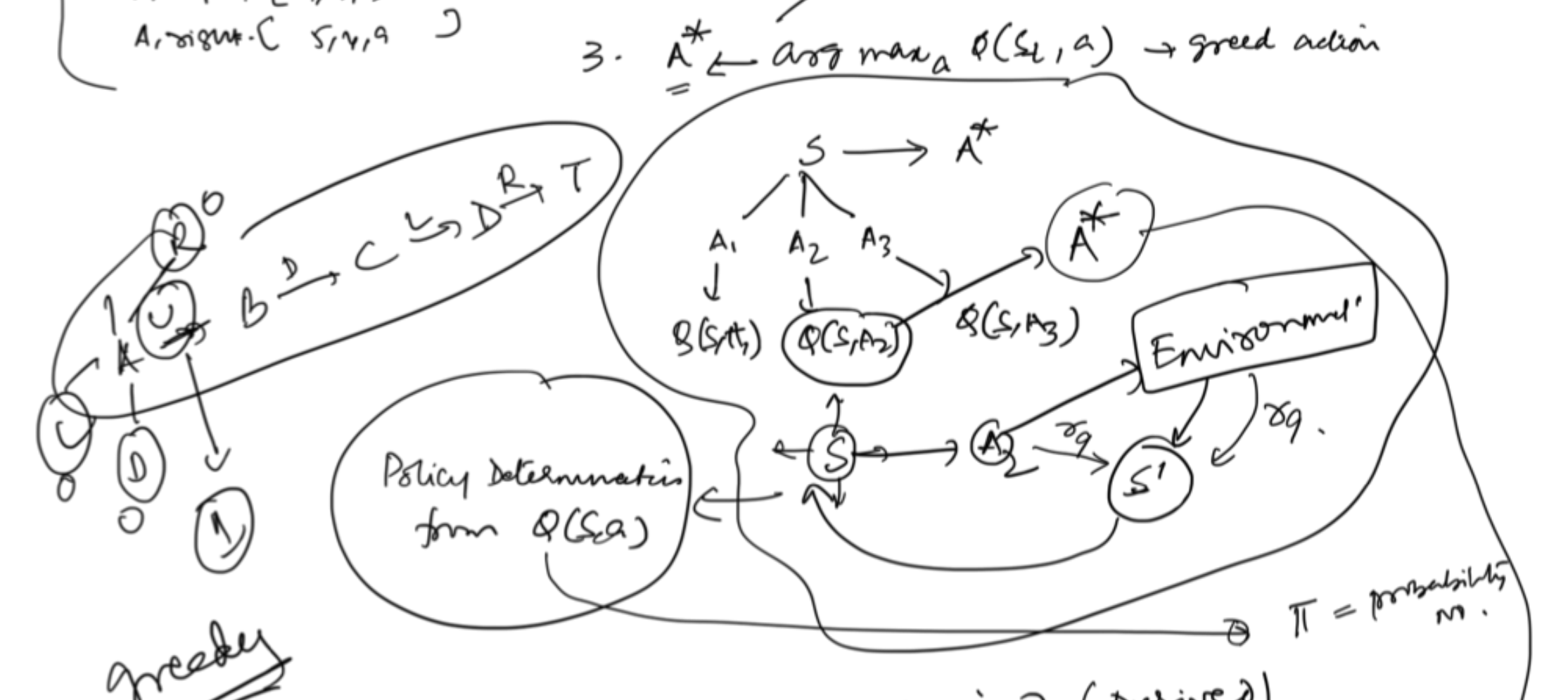
Each visit is counted.

(A, up) = [2,4,3] = Q(A,up) = $\frac{2+4+3}{3}$

A, down = [6,7,8] = Q(A,down) = $\frac{6+7+8}{3}$

A, left = [1,2,3]

A, right = [5,4,9]



→ π = probability m.

4. Policy Determined (Derived)

Prob π determination

5. for all $a \in A(S_t)$

$\pi(a|S_t) \leftarrow \begin{cases} 1-\epsilon+\frac{\epsilon}{|A(S_t)|} & a = A^* \\ \frac{\epsilon}{|A(S_t)|} & a \neq A^* \end{cases}$

greedy

ε greedy

E1 → o-o-o-o . A_t → π

E2 → o-o-o-o . G_t

E3

Probability ←

③ TEMPORAL DIFF

(3.1) What are issues with MC.
- Retrospective method, that waits till the entire episode is over. Q ← update
- What if episode is huge? → Learning is Blocked.
- I know my mistakes only after entire episode. (I → Agent)

MC Loss wrt final Return − Current Return.

TD Loss = next step return − Current Return

(3.2) Algorithm

MC: $V(S_t) \leftarrow V(S_t) + \alpha((G_t) - (V(S_t)))$ Target, Final Return, Present, LOSS, $\ominus$, δ

TD: $V(S_t) \leftarrow V(S_t) + (\alpha)[(R_{t+1} + \gamma V(S_{t+1})) - V(S_t)]$ step, learning rate

INPUT → Policy that needs to be validated

PARAM → $\alpha = (0,1], \gamma, \theta$ → threshold

INITIALIZE $V(S)$ for all S, $V_T = 0$

Online in Realtime, loop for each episode
1. A ← action given my π for S
2. Take action A, observe (R and S') → Env
4. $V(S) \leftarrow V(S) + (\alpha)[\boxed{R + \gamma V(S')} - \boxed{V(S)}]$
5. $S \leftarrow S'$

until S is Terminal.

Target, Current return → $\ominus$ → $\boxed{\delta = \text{Temporal Diff}}$ → Loss function

(α) Learning rate

MC → TARGET was fixed

TD → Target is changing.

(3.3) The changing nature of the target

(10%) — MC used G_t as the Target to learn $V(S_t)$ $\boxed{G_t = R_{t+1} + \gamma G_{t+1}}$

Sampled for that episode → $V(S_t) = V(S_t) + \alpha((G_t) - V(S_t))$ → Target (Back Tracked from last)

Stored

after Running the entire episode

$\boxed{v_\pi(s) \neq V(S)}$ → Stochas., Sample

— $v_\pi(s) = \mathbb{E}_\pi(R_{t+1} + \gamma v_\pi(S_{t+1}) | S_t = S)$

(100%) $P \times \pi \times (R_{t+1} + \gamma v_\pi(S_t))$ → DP

Stochastic

$V(S) \neq Q(S,a)$ → Stochastic, Sampled

1, 5, 6, 3, 4

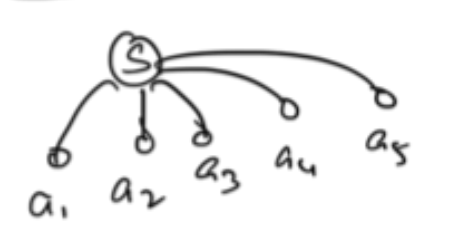
expected value of getting 6 Average

probability of getting 6

$P(1) = \frac{1}{6} \times 1$

$\frac{1}{6}$ = expected value of getting 6

G_t = average of all $Q(S,a)$ of a given S (3 episodes)



$Q(S,a_1) = [Q_1, Q_2, Q_3]$ = avg( ) = $G(S,a)$

$Q(S,a_2) = [Q_5, Q_6, \ ]$

$Q(S,a_3) = [\ \dots\ ]$

$Q(S,a_4) = [\ \dots\ ]$

$Q(S,a_5) = [\ \dots\ ]$

got six?

E1 → 0

E2 → yes

E3 → 0

E4 → 0

E5 → yes.

E100 → ?

$\frac{\text{no. of times I got 6}}{\text{Total no. of Tri.}}$ = Expected = (Avg)   Stochastic

- Estimation in 2 levels. G_t averaging; $V(S_t)$ for $v_\pi(S)$

Mathematics → 0.5 → fair coin prob of H

Engineering → 1000 Episodes ⇒ H; (1) → prob.

R1 sampled ***** (55) average over 55

expected rating of restaurant R1 across 55 samples is 4 → sampled.

Infinity → True Value

expect E(V(S)) 50% = head 50% = Tail

➡ BOTH MC & TD ARE ESTIMATED ALGOS

- In DP, G_t is not computed by actually playing the episode.

$v_\pi(s) = \mathbb{E}_\pi[R_{t+1} + \gamma v_\pi(S_{t+1}) | S_t = s]$

$\sum_a \pi(a|s) \times \sum_{s',r} p(s',r|s,a)\ [r + \gamma v_\pi(s')]$

X in MC

$V(S_t) = V(S_t) + \alpha((G_t) - V(S_t))$ Backward, actual

Forward

TD $V(S_t) \leftarrow V(S_t) + \alpha[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)]$

$\frac{R_{t+1}+R_{t+2}+R_{t+3}}{|t|}$ G_t = Average   Incremental

G_t → Backward = $R_{T-1} + R_{T-2}$

$G_t \leftarrow G_{t+1} \leftarrow G_{t+2} \leftarrow G_{t3}$ 100

E1 (A) → B → C → D → (T) → 0.   R=100

$G_t = R_{t+1} + \gamma(G_{t+1})$

$G_{T-1} = R_{T-1} = 100 + G_T = 100 + 0 = 100$

G_t = average of all returns I would get from a given state.

$G_t = \frac{R_{t+1} + R_{t+2} + R_{t+1} \dots}{n}$ → Backward

MC → Backward

TD → Forward Incremental.

⑤ Example of TD → DRIVING HOME

E1

Sr#	State	Elapsed Time R_{t+1}		Predicted Time TD so $V(S)$		Predicted Total Time G_t
1	Left Office	0	+	30	→	30
2	Reached car. Raining	5	+	35	→	40
3	Exit the freeway	20	+	15	→	35
4	Secondary Road. Behind a big Truck	30	+	10	→	40
5	Enter the home street	40	+	3	→	43
6	Arrive Home	43	+	0	→	43

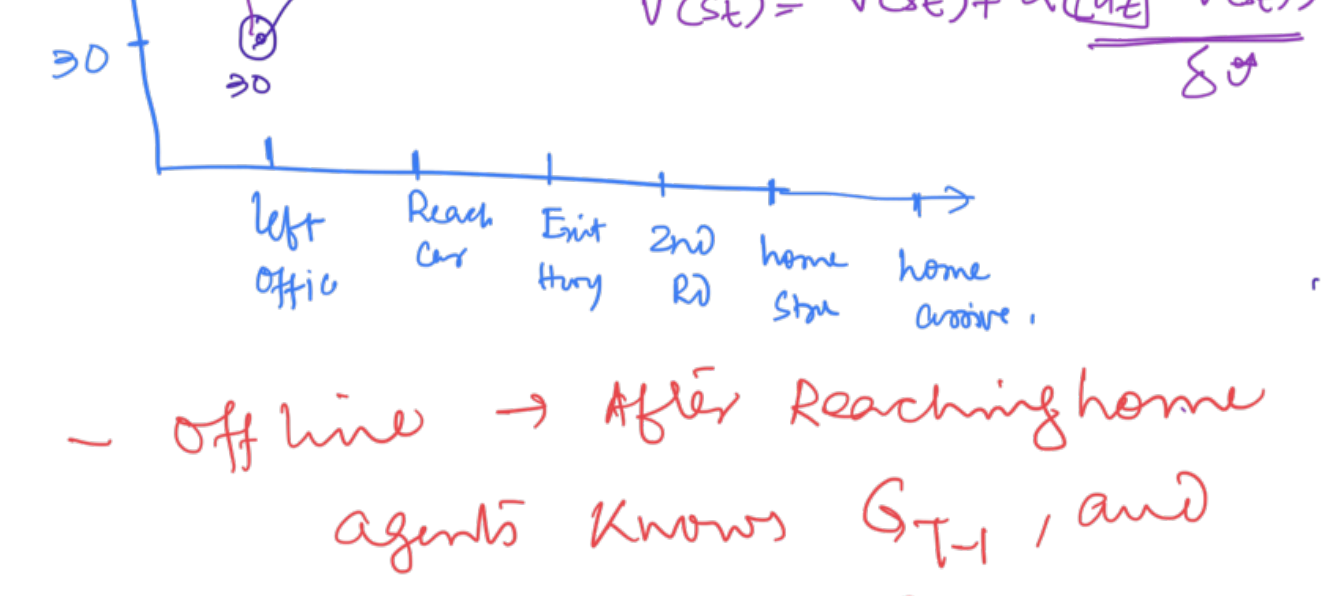
→ Target (fixed)

Actual G_t

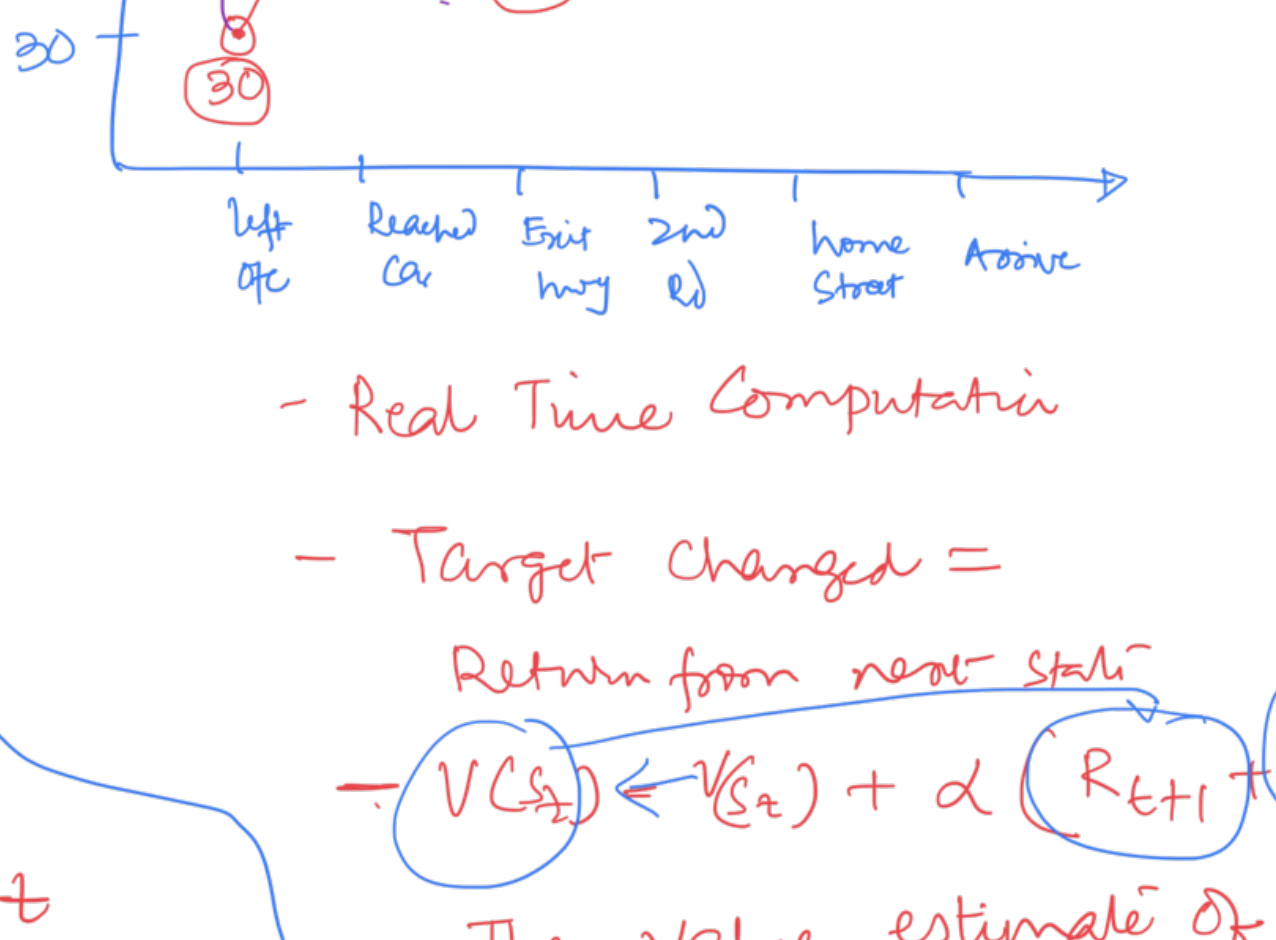
$V_S \leftarrow V_S + \alpha(R_{t+1} + \gamma V(S_t))$

→ retrospect

MONTE CARLO



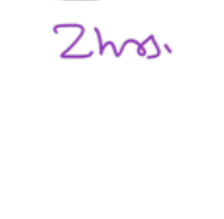
TD



- Off line → After Reaching home agents knows G_{T-1}, and then computes G_t for every step in Backward Retrospection
- Target is Same G_t for every step t
- Value Estimate $V(S_t) \leftarrow V(S_t) + \alpha(G_t - V(S_t))$

NO Looking forward

- Real Time Computation
- Target Changed = Return from next state
- $V(S_t) \leftarrow V(S_t) + \alpha(R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$ Boot Strapping
- The value estimate of step t, is updated in step t+1
- $V(S_t)$ is based on reward and $V(S)$ from (t+1) state.

BOOT STRAPING